

Lesson 4-2

Information Design and User Experience for Websites

Learning Objectives

- Explain how personas and wireframes are used to plan a website before designing its appearance.
- Apply the four principles of design (contrast, repetition, alignment, proximity) and explain the idea of user-centered design.

A website that makes sense to its designer can still confuse its visitors. Good design therefore begins with the user, not the creator. Designers first picture a real user (a persona), sketch where information will go before choosing any colors (a wireframe), and follow four simple principles — contrast, repetition, alignment, and proximity — so the page is clear at a glance. This user-first mindset is called user-centered design.

? Guiding question: How do designers plan a website around its users, and what principles make a page clear and easy to use?

📌 Point!

1. Persona

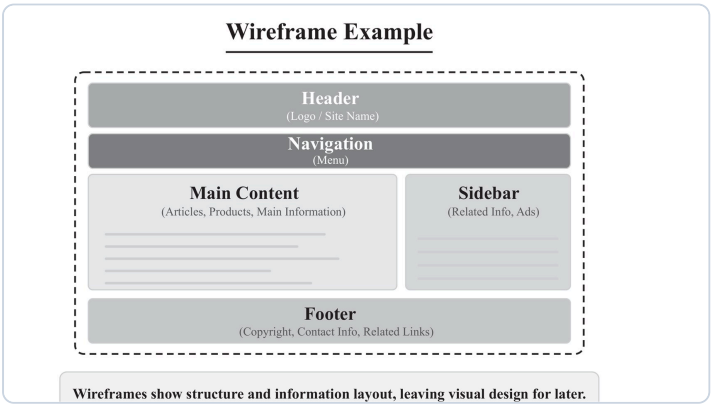
(1) **(Persona)**…a typical user image of those who will use the website, set up as a concrete person. Details such as age, occupation, purpose, and challenges are selected.

(2) By setting up a persona, designers can think specifically about "what this user needs," and the direction of the design becomes less likely to drift.
e.g., Person in their 30s, an office worker, wants to compare products on a smartphone during commute, wants to obtain information in a short time.

2. Wireframe

(1) **(Wireframe)**…a rough blueprint of the layout of a web page. It does not include color or design decoration; it is created to determine the placement and priority of information.

(2) By using a wireframe, "what information to place where" can be considered separately from visual design. If decorations such as colors and fonts are decided first, discussions tend to drift toward personal preferences in appearance, so finalizing the placement and priority of information first makes it easier for stakeholders to reach agreement.



Wireframe Example

Key Fact:
Good web design starts from the user: a persona clarifies who they are, a wireframe fixes what goes where, the CRAP principles make it clear, and user-centered design keeps every choice on the user's side.

Key Concepts:

- persona
- wireframe
- contrast / repetition
- alignment / proximity
- user-centered design

KEY TERMS
persona — a typical user pictured as a concrete person.
wireframe — a layout blueprint before any color or decoration.



A persona captures one target user's goals.

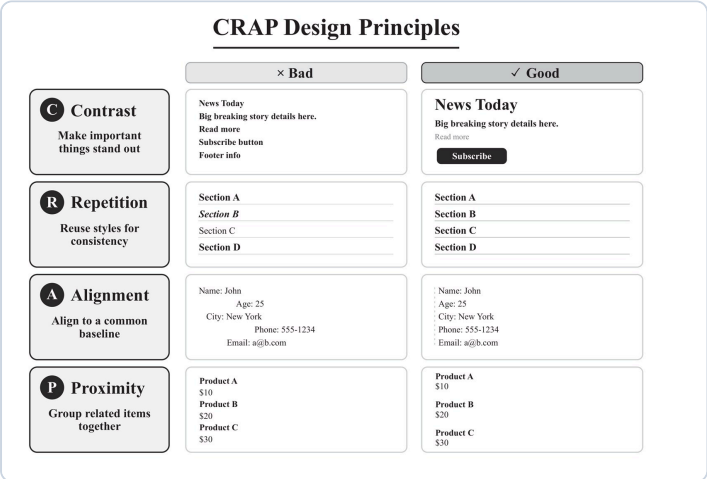


A wireframe grows into a finished design.

3. The Four Principles of Design (CRAP Principles)

(1) Designs that are easy to view and understand follow these four principles.

Principle	Description (purpose)	Specific techniques and examples
(Contrast)	Make the priority of information clear and immediately highlight the most important elements	• Make headings large and bold to differentiate them from body text • Use accent colors for important buttons (e.g., purchase)
(Repetition)	Repeat specific characteristics so that the whole design has consistency, allowing users to navigate without getting lost	• Standardize the color, size, and font of headings on each page • Match the design of bullet point icons
(Alignment)	Align the position of each element to a common baseline, giving order to the entire screen	• Align the left edges of text and images precisely • Make a consistent rule for whether to use center or left alignment
(Proximity)	Place related information close together so that it can be perceived as a single group at a glance	• Group product images, names, and prices together • Leave appropriate spacing (margins) between items



CRAP Design Principles

4. User-Centered Design (UCD)

(1) **(User-Centered Design (UCD))**…a design approach that designs websites and services from the perspective of the user, rather than the convenience of the creator.

(2) In UCD, design begins from “what the user needs” and “in what situation it will be used.” Setting up a persona is one specific way of practicing UCD; by

PAUSE & THINK

Why decide where information goes (wireframe) before choosing colors and fonts?

KEY TERMS

CRAP — contrast, repetition, alignment, proximity.
user-centered design — designing from the user's perspective.

clarifying the user image, design decisions can be kept consistently aligned with the user's perspective.

(3) Improving usability for users and accessibility for a wide range of people are important goals of UCD.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A A persona is a typical user image of those who will use the website, set up as a concrete person.

B A wireframe is a final-version design that includes color and design decoration.

C “Proximity” in the four principles of design means grouping related information close together.

D User-centered design is a design approach that prioritizes the convenience of the creator.

(2) Match each description (a–d) with the most appropriate principle of design from the choices below (A–D).

a Highlight important information and visually indicate priority

b Repeat the same style to give a unified appearance

c Align the positions of elements to give an organized look

d Group related information close together

[Choices] **A** Contrast **B** Repetition **C** Alignment **D** Proximity

✓ Solution

(1)

A Correct definition of a persona. Therefore, ○.

B A wireframe is a blueprint for determining the placement and priority of information, without including color or design decoration. Therefore, ×.

C Correct definition of proximity. Therefore, ○.

D User-centered design is a design approach from the perspective of the user. Therefore, ×.

(2)

a: Highlighting information is contrast. **A**

b: Creating a unified appearance is repetition. **B**

c: Aligning positions is alignment. **C**

d: Grouping related information is proximity. **D**



User-centered design loops around the real user.

PAUSE & THINK

Which principle groups related items together, and which makes the most important item stand out?

Try

1 Answer the following questions.

1. What is the term for a typical user image of those who will use the website, set up as a concrete person?
2. What is the term for a blueprint that shows the placement and priority of information on a web page?
3. What are the four principles of design called when expressed by their English initials?
4. What is the term for the design approach that designs websites and services from the perspective of the user? Also give its three-letter abbreviation.

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate piece of information to include in setting up a persona.

A The user's age and occupation

B The user's purpose for using the website

C The performance of the website's server

D The challenges the user faces

(2) For each of design improvements (a–d), indicate which of the four principles of design (A–D) it corresponds to.

a Making headings large and bold to distinguish them from body text

b Grouping product names, prices, and photos close together

c Aligning the left edges of text and images

d Standardizing the font and color of headings across all pages

A Contrast **B** Repetition **C** Alignment **D** Proximity

Think as an Engineer — Investigate & Decide

A community clinic in your city wants a simple website so patients can find opening hours, services, and directions. You lead the design.

☐ **Start by choosing colors and fonts** ☐ **Start from the user, then design**

1 · Collect data. Ask about eight to ten classmates: “If you opened a community clinic's website, what is the one thing you would look for first?” Record the answers in a tally table.

2 · Analyze your users. Group the answers into user types. Include at least one user who is not a typical smartphone visitor — for example an elderly

EXAM-STYLE QUESTION

Explain why a designer creates a persona and a wireframe before choosing colors and fonts. **[6] Refer to:** the user's needs and reaching agreement with stakeholders.

PAUSE & THINK

For each improvement below, ask: does it highlight, unify, order, or group?

patient with low vision, or someone with no home Internet using a library computer. Write what each type needs most.

3 · Decide from your data. Using your tally, decide the priority order of the four home-page blocks and justify each position from your data, not your taste. Then choose one CRAP principle and say exactly how you would apply it for the user type you added. **In pairs,** compare tables and agree on the top block.

Give evidence for your answer.

Stuck? Start with your tally: which need appeared most? Put that block at the top. Then ask of each block — highlight, unify, order, or group? Pick the CRAP principle that fixes the weakest spot for the user you added.

🌍 In a New Context

A school wants to redesign its cluttered results page so parents can find their child's grades quickly. Name the persona you would define, one wireframe decision, and one CRAP principle you would apply, and say what each one improves.

❓ Lesson Question — Answered

The lesson asked how designers plan a website around its users and what makes a page clear. Design begins from the user, not the creator. A persona pictures a typical user as a concrete person — age, occupation, purpose, and challenges — so the team can focus on what that user needs. A wireframe then fixes the placement and priority of information before any color or decoration, which keeps discussion on structure and helps stakeholders agree. To make the page clear, four principles are applied: contrast to highlight priority, repetition for consistency, alignment for order, and proximity to group related items. All of this is guided by user-centered design (UCD), which starts from what the user needs and aims to improve usability and accessibility. Good design, therefore, is planned from the user inward — who they are, what goes where, and how to make it clear.

KEY TERMS

usability — how easily users can use the site.

accessibility — usable by a wide range of people.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

When designing a website, it is important to view it from the user's perspective. First, create a (a), in which a typical user image is set up as a concrete person, and clarify "what this user needs." Next, create a (b), a blueprint for determining the placement and priority of information. In design, to improve viewability and clarity, the (c)—contrast, repetition, alignment, and proximity—should be kept in mind.

1. Fill in the blanks (a)–(c).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT included in the four principles of design (CRAP principles).

A Contrast
B Encryption
C Alignment
D Proximity

(2) From the following options (A–D), choose the one that most appropriately describes the concept of user-centered design.

A Prioritize the design that the developer finds easy to use.
B Place the highest importance on visual design.
C Begin the design from what the user needs.
D Use as many of the latest technologies as possible.

⭐ REFLECT & CHALLENGE

Reflect: On a website you find confusing, which CRAP principle seems most broken, and why? **Challenge:** Your clinic can fund only one accessibility improvement this year. For the low-vision or no-Internet user you identified, choose one CRAP principle to apply — and say what you would give up (a decoration or feature) to make room for it, and why that trade-off serves the user better.

⭐ Key takeaway

Remember:
Persona (who) → wireframe (where) → CRAP (make it clear) → UCD (decide for the user). Design from the user inward.